competing consolidator's Form CC. The Commission has considered this comment and is amending its ongoing estimate after a review of amendments of the fee schedules of the SROs.

(iii)Adopted Estimates – Ongoing Burden and Costs

As described above,[1421] the Commission is amending its ongoing estimate that a competing consolidator will file two amendments per year. The Commission now estimates that a competing consolidator will file six amendments per year. The Commission believes a competing consolidator will file five amendments a year, plus the annual report, for a total of six amendments per year. The Commission believes this estimate is reasonable based upon a review of amendments of the fee schedules of the SROs. Because the Commission believes that a competing consolidator will incur an ongoing burden of 0.25 hours per amendment to ensure that it has posted the correct direct URL hyperlink to the Commission's website on its own website,[1422] the Commission now estimates that a competing consolidator will incur a total of 1.5 hours per year to ensure that it has posted the correct direct URL hyperlink to the Commission's website,[1423] for an aggregate annual burden of approximately 12 hours for all competing

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1421 See supra Section IV.D.1(b)(iii).

1422 See supra note 1418.

1423 The Commission bases this estimate on a full-time Programmer Analyst spending approximately 0.25 hours to check the Commission's website when the competing consolidator submits an amendment to effective Form CCs to ensure that the Commission's direct URL hyperlink that the competing consolidator has posted to its own website remains valid. Since the Commission estimates that a competing consolidator would file six amendments per year, the Commission estimates that each competing consolidator would incur a burden of 1.5 hours per year. (0.25 hours) x (6 amendments per year) = 1.5 hours per year to check the URL hyperlink. The Commission estimated the monetized annual burden for this requirement to be $361.50. The Commission derives this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by

consolidators to do so.[1424] The Commission is adopting the ongoing burden estimate as amended.

3. Competing Consolidator Duties and Data Collection

Rules 614(d)(1) through (d)(3) require competing consolidators to collect from the SROs quotation and transaction information for NMS stocks, calculate and generate a consolidated market data product, and make the consolidated market data product available to subscribers on a consolidated basis on terms that are not unreasonably discriminatory. Rule 614(d)(4) requires competing consolidators to timestamp the information with respect to quotations and transactions in NMS stocks that they collect from the SROs pursuant to Rule 614(d)(1) upon receipt, upon receipt by the aggregation mechanism, and upon dissemination to subscribers. The Commission estimated that five types of entities would register to become competing consolidators and would have to build systems, or modify existing systems, to comply with Rules 614(d)(1) through (d)(4): (1) market data aggregation firms, (2) broker-dealers that currently aggregate market data for internal uses, (3) the existing exclusive SIPs (CTA/CQ and Nasdaq UTP SIPs), (4) entities that would be entering the market data aggregation business for the first time ("new entrants"), and (5) SROs. The Commission estimated that, apart from the SRO category, two respondents

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5.35 to account for bonuses, firm size, employee benefits, and overhead: Programmer Analyst at $241 for 1.5 hours = 1.5 annual burden hours per competing consolidator and $361.50.

1424 The Commission estimates the monetized aggregate annual burden for this requirement to be $2,892. The Commission derives this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Programmer Analyst at $241 for 1.5 hours) x (8 competing consolidators)] = 12 annual burden hours across the competing consolidators and $2,892.

from each category would register to become a competing consolidator; the Commission estimated that four SROs would register to become competing consolidators.

(a) Comments on Initial Burden and Costs and Annual Burden and Costs Generally

The Commission received two comments that believed the Commission's initial or ongoing burdens and costs associated with the operation of competing consolidators were too low.[1425] One commenter said the Commission's estimated initial and ongoing costs associated with competing consolidators should be comparable to those of the CAT.[1426] The Commission considered this comment and disagrees with its assessment because the CAT is a different system in function and differs greatly in scope than the systems to be used by competing consolidators. Unlike the systems to be operated by competing consolidators, which would collect trade and quote information in NMS stocks from the SROs, and consolidate and disseminate such information to subscribers, the CAT must collect information for the entire lifecycle of an order (receipt/origination, routing, receipt of a routed order, modification or cancellation, and execution), in both NMS stocks and options from SROs as well as broker-dealers, and consolidate and store such information in a queryable database made available to regulators.

The other commenter stated that the cost that NYSE incurred to build its "NMS network" inside one data center to provide access to SIAC's NMS feeds "was substantially greater than the

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1425 Data Boiler Letter I at 46; Data Boiler Letter II at 1; IDS Letter I at 13.

1426 Data Boiler Letter I at 46; Data Boiler Letter II at 1.

Commission's estimation for networks that would extend to four data centers."[1427] The commenter said that NYSE's capital expenditure costs to build the NMS network were estimated to be $3.8 million, with ongoing costs of $215,000 per year.[1428] The Commission considered this comment and believes the NMS network costs are informative but are not directly applicable to the costs to be incurred by competing consolidators to build or upgrade systems to comply with Rules 614(d)(1) through (d)(4) because the NMS network is not a system that consolidates market data and its costs include the transmission of options information, which competing consolidators would not be collecting, consolidating, or disseminating. However, upon further consideration, the Commission believes that it is likely that competing consolidators would incur higher technology-related burden hours and external costs associated with building as well as operating systems to collect, consolidate, and disseminate consolidated market data than the Commission estimated in the proposal. The Commission is increasing its estimates accordingly.

As adopted, Rule 614(d)(1) through (d)(3) does not require competing consolidators to sell a full consolidated market data product.[1429] Competing consolidators that decide to offer a limited consolidated market data product may incur fewer burden hours and costs to build and maintain a system that does not have to aggregate and disseminate a full consolidated market data product. However, the Commission believes that there will continue to be demand for a full consolidated market data product, which will incentivize some competing consolidators to meet this demand.[1430] Therefore, the Commission is not reducing its estimated burden hours and external costs for competing consolidators to implement and maintain systems to comply with

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1427 IDS Letter I at 13, n.38.

1428 Id.

1429 See supra Section III.C.8(a)(ii). See also supra Sections II.B.2; III.C.1(b).

1430 See supra notes 878–880 and accompanying text. See also infra Section V.C.1(c).

Rules 614(d)(1) through (d)(4) to reflect this change to the data they must make available. The Commission acknowledges that these burden hours and external costs reflect an upper bound and as incurred may be lower than these estimates for those competing consolidators that sell a limited consolidated market data product.

(b) Initial Burden and Costs for Market Data Aggregation Firms

(i) Proposed Estimates – Initial Burden and Costs

In the Proposing Release, the Commission estimated that each market data aggregation firm would incur 900 initial burden hours[1431] and $206,250 in external costs[1432] to modify its systems to comply with Rules 614(d)(1) through (d)(4). Additionally, the Commission estimated that an existing market data aggregator would incur initial external costs of $14,000 to purchase market data from the SROs,[1433] and an additional initial external cost of $194,000 to co-locate at four exchange data centers,[1434] for a total initial external cost of $414,250 per existing market

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1431 The Commission estimated the monetized initial burden for this requirement to be $293,750. Based on discussions with a market participant, the Commission reached the following estimates: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 6 months (900 burden hours) to upgrade existing systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1432 This estimate was based on discussions with a market participant and the Commission's understanding of hardware costs.

1433 The Commission used the monthly market data access and redistribution fees currently charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).

1434 This estimate was based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. The Commission

data aggregator,[1435] and an aggregate estimate for two market data aggregators of 1,800 initial burden hours[1436] and $828,500 in initial external costs.[1437]

(ii) Comments/Responses on Initial Burden and Costs

In response to the commenter that believed that the estimated costs incurred by potential competing consolidators to build or upgrade systems to comply with proposed Rules 614(d)(1) through (d)(4) should be increased,[1438] the Commission is modifying its initial burden and cost estimates for market data aggregators, as discussed below.

(iii)Adopted Estimates – Initial Burden and Costs

The Commission is increasing its estimated initial costs and associated burden hours for market data aggregators to modify their systems to comply with Rules 614(d)(1) through (d)(4). The Commission preliminarily believed that market data aggregators would not have to extensively modify their systems to comply with Rules 614(d)(1) through (d)(4) because the

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described that the market data aggregators would already be co-located at the four exchange data centers, which could lower the estimate. See NYSE Price List 2020, available at https://www.nyse.com/publicdocs/nyse/markets/nyse/NYSE_Price_List.pdf (last accessed Nov. 27, 2020).

1435 $414,250 = [($206,250 in initial external costs to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 for the first month of market data costs) + ($194,000 in initial co-location costs at four exchange data centers)].

1436 The Commission estimated the monetized initial burden for this requirement to be $587,500. Based on discussions with a market participant, the Commission reached the following estimates: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] x [(2 market data aggregation firms)] = 1,800 initial burden hours across the market data aggregation firms.

1437 The Commission estimated that the market data aggregation firms would incur the following initial external costs: [($206,250 to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to co-locate within four exchange data centers)] x [(2 market data aggregation firms)] = $828,500.

1438 IDS Letter I at 13, n.38.

systems used by these firms already collect, consolidate, and disseminate more extensive proprietary market data than the data that is provided by the exclusive SIPs. However, the Commission now understands that these are small firms for which scaling out their hardware and personnel needs will be a significant undertaking. Most of these firms would have to spend substantial time coding for the new technical changes and would likely not have all of the components required to comply with Rules 614(d)(1) through (d)(4). Additionally, the Commission initially believed that competing consolidators would build aggregation systems in a single data center; however, the Commission now believes that competing consolidators may build systems for aggregating data in more than one data center. The Commission believes market data aggregators would likely incur external costs greater than the Commission's estimate to buy new technology (for example, hardware and network infrastructure).

The Commission is increasing its estimated burden hours for Sr. Programmers and Sr. Systems Analysts employed by market data aggregation firms by three times. The Commission initially believed that competing consolidators would build aggregation systems in a single data center; however, the Commission now believes that competing consolidators may build systems for aggregating data in more than one data center, so the Commission is increasing the hours for these technical job categories by three times because competing consolidators may potentially build aggregation systems in three data centers. The Commission is also increasing its estimated external costs to be incurred by market data aggregation firms to purchase new technology to upgrade their systems to comply with Rules 614(d)(1) through (d)(4) by three times for the same reason. The Commission estimates that each market data aggregation firm would incur 2,200

initial burden hours to modify its systems to comply with Rules 614(d)(1) through (d)(4),[1439] and initial external costs of $618,750 to purchase the necessary technology to effect such modifications,[1440] $194,000 to establish co-location connectivity to the exchange data centers,[1441] and $14,000 to purchase market data from the exchanges,[1442] for a total external cost to each market data aggregator of $826,750.[1443] The Commission estimates that the total initial burden

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1439 The Commission estimated the monetized initial burden for this requirement to be $697,150. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 1,050 hours) + (Sr. Systems Analyst at $285/hour for 900 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 2,200 initial burden hours to upgrade existing systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1440 This estimate was originally based on discussions with a market participant and the Commission's understanding of hardware costs. The Commission has increased this estimated cost by three times because the Commission believes competing consolidators may potentially build aggregation systems in three data centers.

1441 This estimate is based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. The Commission believes that the market data aggregators would already be co-located at the four exchange data centers, which may lower this estimate. See NYSE Price List 2020, supra note 1434.

1442 As it did in the Proposing Release, the Commission is using the monthly market data access and redistribution fees charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).

1443 The Commission estimated that each market data aggregation firm would incur the following initial external costs: [($618,750 to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] = $826,750.

hours for two market data aggregators would be 4,400 burden hours,[1444] and that total initial external costs would be $1,653,500 for two market data aggregators to modify their systems to comply with Rules 614(d)(1) through (d)(4).[1445]

(c) Initial Burden and Costs for Broker-Dealers that Aggregate Market Data

(i) Proposed Estimates – Initial Burden and Costs

In the Proposing Release, the Commission estimated that each broker-dealer that aggregates market data for internal uses that chooses to become a competing consolidator would incur burden hours to upgrade its systems to comply with Rules 614(d)(1) through (d)(4) as well as external costs associated with such upgrades, including co-location fees at the exchange data centers and the cost of market data. Specifically, the Commission estimated that each broker-

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1444 The Commission estimated the monetized initial burden for this requirement to be $1,394,300. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 1,050 hours) + (Sr. Systems Analyst at $285/hour for 900 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] x [(2 market data aggregation firms)] = 4,400 initial burden hours across the market data aggregation firms.

1445 The Commission estimated that market data aggregation firms would incur the following initial external costs: [($618,750) to modify systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] x [(2 market data aggregation firms)] = $1,653,500.

dealer would incur 900 initial burden hours[1446] and $206,250 in external costs[1447] to modify its systems to comply with Rules 614(d)(1) through (d)(4). Additionally, the Commission estimated that a broker-dealer would incur initial external costs of $14,000 to purchase market data from the SROs,[1448] and an additional initial external cost of $194,000 to co-locate itself at four

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1446 The Commission estimated the monetized initial burden for this requirement to be $293,750. The Commission reached the following hourly estimates: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 6 months (900 burden hours) to upgrade existing systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on discussions with a market participant and per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for a 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1447 This estimate was based on discussions with a market participant and the Commission's understanding of hardware costs.

1448 The Commission used the monthly market data access and redistribution fees currently charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).